

FLUIDIC SELF-ASSEMBLY TRANSFER TECHNOLOGY FOR MICRO-LED DISPLAY

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ABSTRACT

In this paper, we present a demonstration of high yield Fluidic Self-Assembly (FSA) technology to transfer gallium nitride (GaN) microchips which can be used for Micro-LED display. The low melting point alloy on the substrate and the metal electrode of the chip were assembled in heated solution by a simple shaking motion. More than 19,000 blue GaN microchips with 45 μ m in diameter were precisely assembled at 99.90% yield within 1 min. At the chip sizes below 50 μ m, this dramatic improvement in assembly yield was achieved by using a new assembly solution and chip designs. This low cost and fast demonstration has proven that FSA is a suitable mass transfer technology which is applicable to Micro-LED display after some further development.

KEYWORDS

Fluidic Self-Assembly (FSA), low cost, mass transfer technology, GaN microchip, Micro-LED display.

INTRODUCTION

Micro-LED display, arrays of sub-100 μ m inorganic LED which individually operates as a pixel, has been considered as a next generation display. For Micro-LED display, gallium nitride (GaN) chips are separately grown on the donor wafers and transferred to the substrate. Therefore, the mass transfer technology, transferring millions of chips with low cost and high-throughput, is necessary. Unfortunately, currently used pick-and-place or transfer printing is not suitable for Micro-LED display as the display size and its resolution increase. The process of pick-and-place is picking up the chips from one area and then placing them to the other. This serial process requires long working hours and precise alignment. Transfer printing is picking up the components with a stamp material from donors and transferring them on the target substrate. It also needs many iterations and precise align accuracy.

On the other hand, FSA has a benefit in mass transfer because each component can be assembled simultaneously under the certain environment. Its parallel process has a great advantage in working time. Although FSA has this strong point, it could not be used as a Micro-LED display manufacturing technology yet because of its low assembly yield. The yield must be improved significantly in order to apply FSA technology to the industry of Micro-LED display.

Contributions

- Improved wettability of molten alloy to gold

(Au) surface by adding surfactant in the assembly solution

- Optimized the chip design for one-on-one assembly and align accuracy
- Assembled 19,663 circular GaN chips with a diameter of 45 μ m in 1 min
- Achieved 99.90% assembly yield with a simple shaking FSA

SYSTEM DESCRIPTION

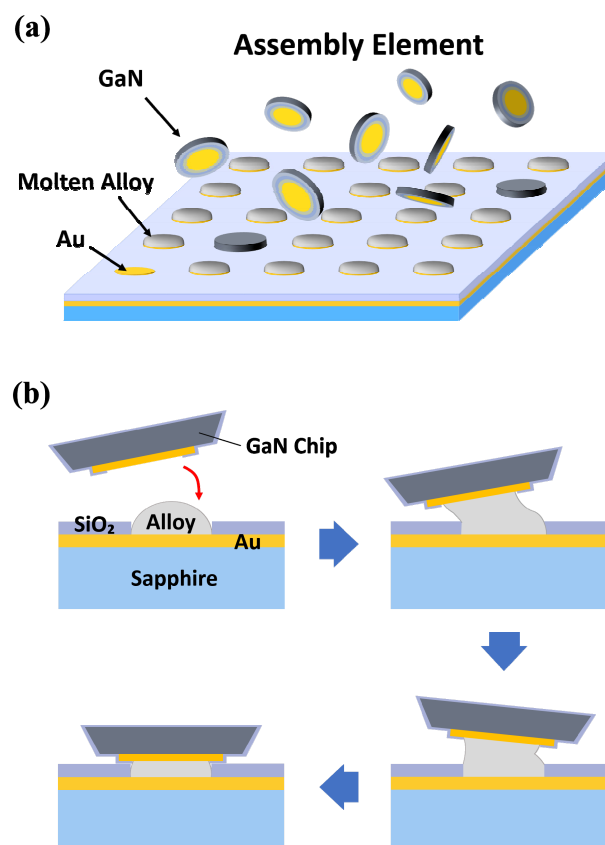


Figure 1 : Schematic illustration for the mechanism of FSA. (a) schematic illustration of the assembly elements (b) description of capillary force driven assembly.

Here, we assembled 19,663 GaN chips using FSA and achieved a yield of 99.90%. By simply shaking a heated glass vial containing assembly solution, the target substrate and GaN chips, low melting point alloy on the substrate and the metal electrode of the chip were assembled by capillary force (Figure 1). Although the shaking FSA method has been already implemented by other researchers [1], they assembled only a few hundred chips of several

hundred micrometers scale and the yield was less than 99%. To increase the assembly yield of sub-hundred micrometer sized chips, we used F108 surfactants. By adding the surfactant to the assembly solution, the surfaces of Au electrode on the chips were modified. Surface energy of the modified Au was increased so that the wettability with molten alloy was improved. This caused the assembly yield to be significantly increased.

State-of-the-art

In most cases of Self-Assembly, the chips were greater than 100um and cube like shape [1][2][3][6][7][8]. One of the most important factors in Self-Assembly is setting the conditions for the chips to be positioned on the binding sites. There are many methods to guide and move the chips to the sites. For example, chips can be sprinkled directly over the sites, rolled off by gravity, moved along the interface between two liquids, or guided with magnetic actuation.

However, most methods were used to assemble chips larger than 100um, and the yields were less than 99%. Even though the technology of pulling up the chips along the interface of two liquids only has succeeded to assemble the chips below 100um scale, it still could not exceed 99% yield [7]. The state-of-the-art Self-Assembly is described in Table 1.

Group	Chip Size (um)	Shape	Assembled #	Time (sec)	Yield
[1]	280	cube	1560	180	98%
[2]	2000 x 2000 x 100	box	81	70	100%
[3]	790	cube	1000	140	99.8%
[4]	100	variety	10000	1500	97%
[5]	Φ : 2000/h: 100	Disk	90	60	100%
[6]	1000 x 1000 x 500	box	2500	300	97%
[7]	20 ~ 60	cube	62500	45	98%
[8]	500	cube	1524	3600	99.6%
Our Work	Φ: 45/h: 5	disk	19663	60	99.90%

Table 1 : State-of-the-art Self-Assembly is summarized. Chip size, shape of chip, number of assembled chip, time taken for assembly, and assembly yield are shown. Our work is described at the last row.

EXPERIMENT

Substrate preparation

Au layer of 100nm thickness was deposited on a 2-inch sapphire wafer as a common electrode using an e-beam. On the top of the layer, SiO₂ of 50nm thickness was covered. Then, circular holes with a diameter of 30um were etched in 243 rows and 81 columns until Au surface is exposed. The pitches were 120um and 360um, respectively. The opened Au areas on the substrate was then coated with an alloy that has a melting point of 60°C through dip soldering process (Figure 2 a). The alloy pot for soldering contained slightly acidic water (pH 3) on top of the molten alloy to remove the oxide layers on the alloy surface. The pot was heated to 75°C to melt the alloy. When the alloy was molten, the substrate was put into it for 30 seconds then pulled out.

Optimizing the design of GaN chip

We designed GaN chip to maximize the assembly yield

and accuracy. Usually, the chip has a rectangular shape because it is easy to make. However, in FSA, it is hard to make a uniform array with rectangular shaped chips because they are assembled differently according to the assembly angle in x and y axis. For this reason, we decided to use a circular design which is angle-independent. N-type GaN, InGaN, and P-type GaN layers were grown on the sapphire wafer, and Au electrode layer with a thickness of 100nm were fabricated on the P-type GaN. The GaN chip has a circular shape with a diameter of 45um and the Au electrode was designed to be smaller than the chip itself, having a diameter of 30um (Figure 2 b). The reason why the electrode being smaller than the chip was to intend each chip being assembled one-on-one at a binding site.

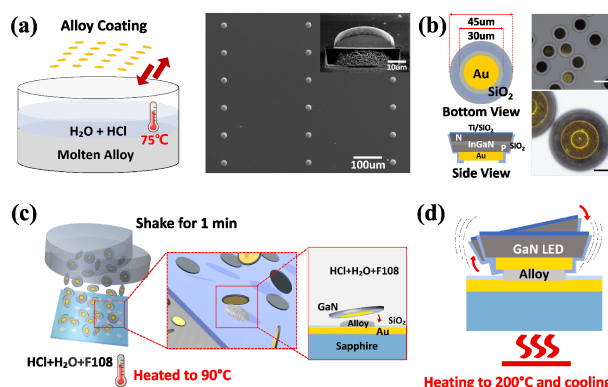


Figure 2 : Schematic illustration for the process of FSA. (a) preparation and SEM images of the substrate (b) bottom and side view of the chip structure and microscopic images; scale bar white=50um, black=10um, (c) description of FSA process (d) adjustment through heat reflow after FSA.

Process of Fluidic Self-Assembly

The assembly solution was prepared by adding 2wt% F108 surfactant and a few drops of hydrochloric acid (HCl) to deionized water (DI water) to make pH 3. We put about 3 millions of GaN chips, alloy coated substrate, and the assembly solution into a glass vial. Then, the vial was heated until the solution reached 90°C and sealed with a Teflon lid after reaching the target temperature. The vial was heated above the melting point of the alloy in consideration of natural cooling. The sealed vial was then shaken for about 60 seconds for assembly (Figure 2 c). After shaking, the substrate was pulled out and rinsed with Isopropyl alcohol (IPA) and DI water in order.

Heat reflow after assembly

After the assembly, we conducted heat reflow process to revise the chips that were assembled on the tilt. The process was performed in 200°C furnace for 5 min (Figure 2 d). By the heat reflow and slow cooling, the tilted chips were adjusted to flat.

RESULT

Assembled number and yield

Total number of assembly site was 19683 (243 by 81). As a best result, 19663 undamaged chips and 9 defective chips were assembled one-on-one on the binding sites. Some chips were cracked or broken either during the assembly

process or chip separation from the donor sapphire wafer. In terms of assembly yield, the demonstration reached 99.94% regardless of the defects (Figure 3). The yield was 99.90% even excluding the damaged chips.

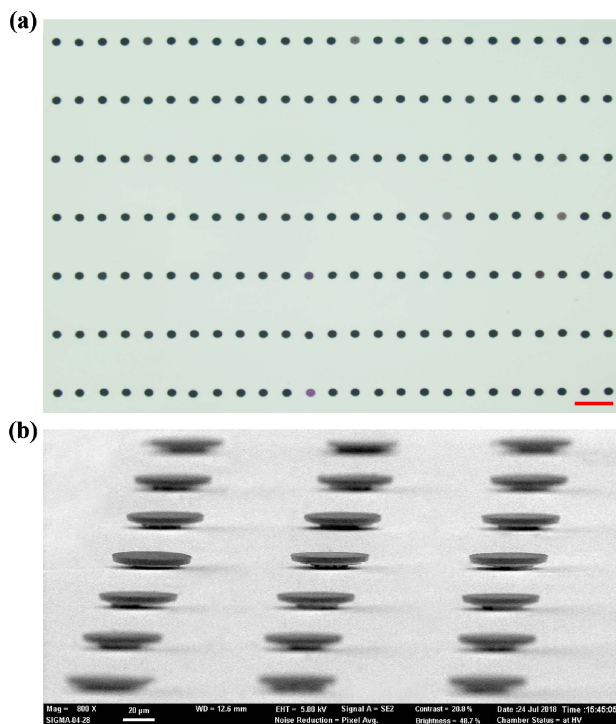


Figure 3 : Images of assembled GaN chips on the substrate. (a) microscopic image of assembled chips on the substrate; scale bar red=200um (b) SEM image of assembled chips.

CONCLUSIONS

We successfully assembled GaN chips in uniformly spaced array shape through FSA technology. The best yield was 99.90%. We improved wettability of molten alloy to the Au electrode on the chip by adding F108 surfactant to the assembly solution. Optimization of the design of GaN chip enabled precise one-on-one assembly. This simple and low cost assembly method has a chance to be applied to the mass transfer technology for Micro-LED display. However, this method is immature yet and still has to be developed for the commercial usage since it uses low melting temperature alloy. The alloy can melt when the heat above 60°C is applied to the array, so that the assembled chips become inoperable. In addition, the yield needs to be increased even higher for the display industry. Further research should be followed for better durability and higher assembly yield.

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